

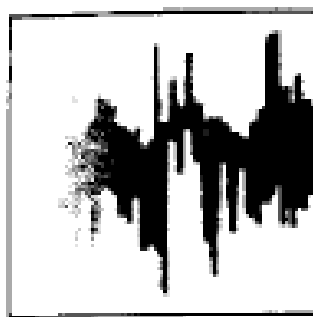


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3

The Z -Transform and Its Application to the Analysis of LTI Systems

Transform techniques are an important tool in the analysis of signals and linear time-invariant (LTI) systems. In this chapter we introduce the z -transform, develop its properties, and demonstrate its importance in the analysis and characterization of linear time-invariant systems.

The z -transform plays the same role in the analysis of discrete-time signals and LTI systems as the Laplace transform does in the analysis of continuous-time signals and LTI systems. For example, we shall see that in the z -domain (complex z -plane) the convolution of two time-domain signals is equivalent to multiplication of their corresponding z -transforms. This property greatly simplifies the analysis of the response of an LTI system to various signals. In addition, the z -transform provides us with a means of characterizing an LTI system, and its response to various signals, by its pole-zero locations.

3.1 THE Z-TRANSFORM

3.1.1 The Direct z-Transform

The z -transform of a discrete-time signal $x(n)$ is defined as the power series

$$X(z) \equiv \sum_{n=-\infty}^{\infty} x(n)z^{-n} \quad (3.1.1)$$

where z is a complex variable. The relation (3.1.1) is sometimes called the *direct z-transform* because it transforms the time-domain signal $x(n)$ into its complex-plane representation $X(z)$. The inverse procedure [i.e., obtaining $x(n)$ from $X(z)$] is called the *inverse z-transform* and is examined briefly in Section 3.1.2 and in more detail in Section 3.4.

For convenience, the z -transform of a signal $x(n)$ is denoted by

$$X(z) \equiv Z\{x(n)\} \quad (3.1.2)$$

whereas the relationship between $x(n)$ and $X(z)$ is indicated by

$$x(n) \xleftrightarrow{z} X(z) \quad (3.1.3)$$

Since the z -transform is an infinite power series, it exists only for those values of z for which this series converges. The *region of convergence* (ROC) of $X(z)$ is the set of all values of z for which $X(z)$ attains a finite value. Thus any time we cite a z -transform we should also indicate its ROC.

We illustrate these concepts by some simple examples.

Example 3.1.1

Determine the z -transforms of the following *finite-duration* signals.

(a) $x_1(n) = \{1, 2, 5, 7, 0, 1\}$

(b) $x_2(n) = \{1, 2, 5, 7, 0, 1\}$

Solution From definition (3.1.1), we have

(a) $X_1(z) = 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-4}$, ROC: entire z -plane except $z = 0$

(b) $X_2(z) = z^2 + 2z + 5 + 7z^{-1} + z^{-3}$, ROC: entire z -plane except $z = 0$ and $z = \infty$

From this example it is easily seen that the ROC of a *finite-duration signal* is the entire z -plane, except possibly the points $z = 0$ and/or $z = \infty$. These points are excluded, because $z^k (k > 0)$ becomes unbounded for $z = \infty$ and $z^{-k} (k > 0)$ becomes unbounded for $z = 0$.

From a mathematical point of view the z -transform is simply an alternative representation of a signal. This is nicely illustrated in Example 3.1.1, where we see that the coefficient of z^{-n} , in a given transform, is the value of the signal at time n . In other words, the exponent of z contains the time information we need to identify the samples of the signal.

In many cases we can express the sum of the finite or infinite series for the z -transform in a closed-form expression. In such cases the z -transform offers a compact alternative representation of the signal.

Example 3.1.2

Determine the z -transform of the signal

$$x(n) = \left(\frac{1}{2}\right)^n u(n)$$

Solution The signal $x(n]$ consists of an infinite number of nonzero values

$$x(n) = \{1, (\tfrac{1}{2}), (\tfrac{1}{2})^2, (\tfrac{1}{2})^3, \dots, (\tfrac{1}{2})^n, \dots\}$$

The z -transform of $x(n]$ is the infinite power series

$$\begin{aligned} X(z) &= 1 + \tfrac{1}{2}z^{-1} + (\tfrac{1}{2})^2z^{-2} + (\tfrac{1}{2})^nz^{-n} + \dots \\ &= \sum_{n=0}^{\infty} (\tfrac{1}{2})^nz^{-n} = \sum_{n=0}^{\infty} (\tfrac{1}{2}z^{-1})^n \end{aligned}$$

This is an infinite geometric series. We recall that

$$1 + A + A^2 + A^3 + \dots = \frac{1}{1 - A} \quad \text{if } |A| < 1$$

Consequently, for $|\frac{1}{2}z^{-1}| < 1$, or equivalently, for $|z| > \frac{1}{2}$, $X(z)$ converges to

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} \quad \text{ROC: } |z| > \frac{1}{2}$$

We see that in this case, the z -transform provides a compact alternative representation of the signal $x(n)$.

Let us express the complex variable z in polar form as

$$z = re^{j\theta} \tag{3.1.4}$$

where $r = |z|$ and $\theta = \angle z$. Then $X(z)$ can be expressed as

$$X(z)|_{z=re^{j\theta}} = \sum_{n=-\infty}^{\infty} x(n)r^{-n}e^{-j\theta n}$$

The following examples illustrate these important concepts.

Example 3.1.3

Determine the z -transform of the signal

$$x(n) = \alpha^n u(n) = \begin{cases} \alpha^n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

Solution From the definition (3.1.1) we have

$$X(z) = \sum_{n=0}^{\infty} \alpha^n z^{-n} = \sum_{n=0}^{\infty} (\alpha z^{-1})^n$$

If $|\alpha z^{-1}| < 1$ or equivalently, $|z| > |\alpha|$, this power series converges to $1/(1 - \alpha z^{-1})$.

Thus we have the z -transform pair

$$x(n) = \alpha^n u(n) \xleftrightarrow{z} X(z) = \frac{1}{1 - \alpha z^{-1}} \quad \text{ROC: } |z| > |\alpha| \quad (3.1.7)$$

The ROC is the exterior of a circle having radius $|\alpha|$. Figure 3.2 shows a graph of the signal $x(n)$ and its corresponding ROC. Note that, in general, α need not be real.

If we set $\alpha = 1$ in (3.1.7), we obtain the z -transform of the unit step signal

$$x(n) = u(n) \xleftrightarrow{z} X(z) = \frac{1}{1 - z^{-1}} \quad \text{ROC: } |z| > 1 \quad (3.1.8)$$

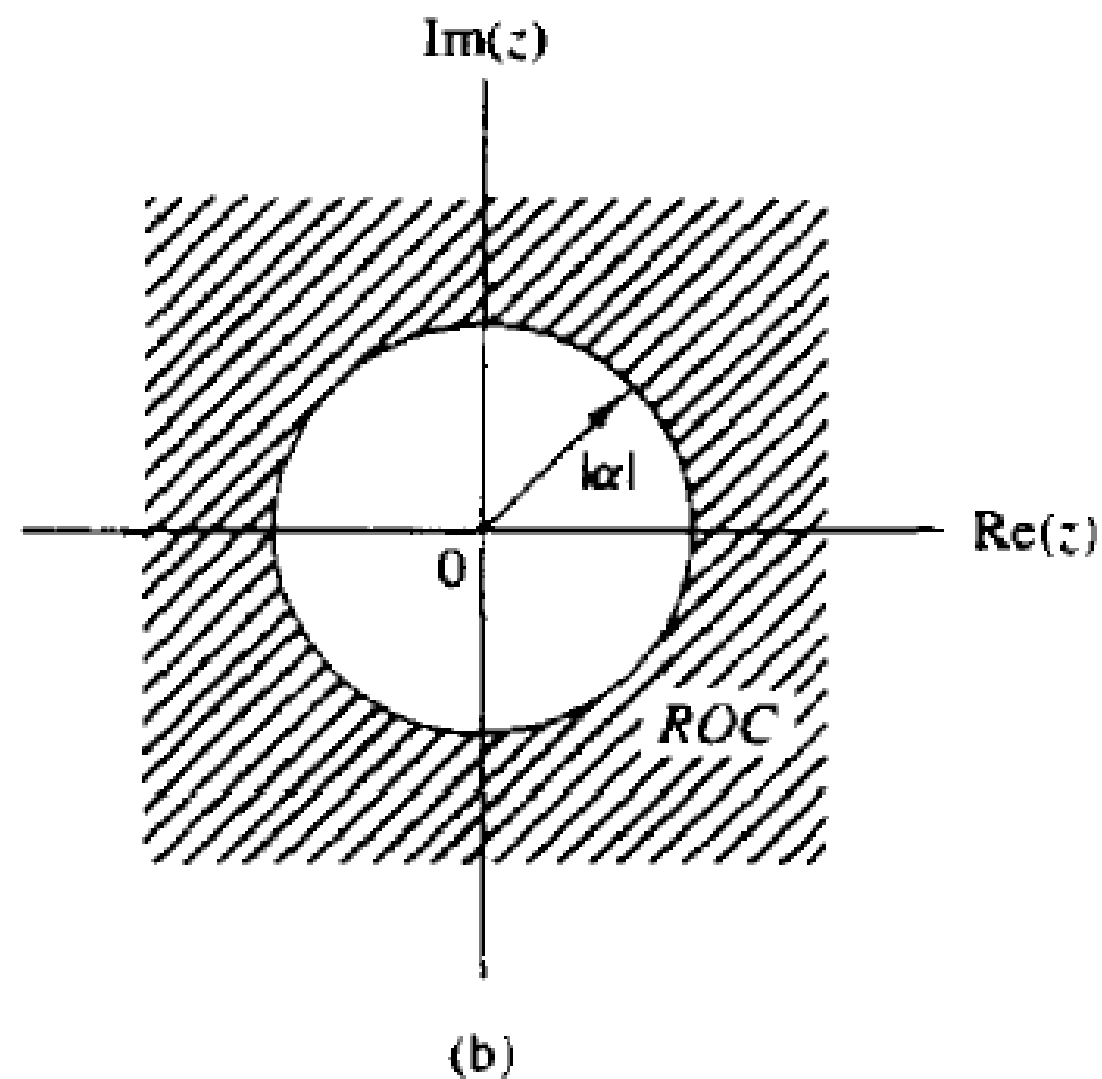
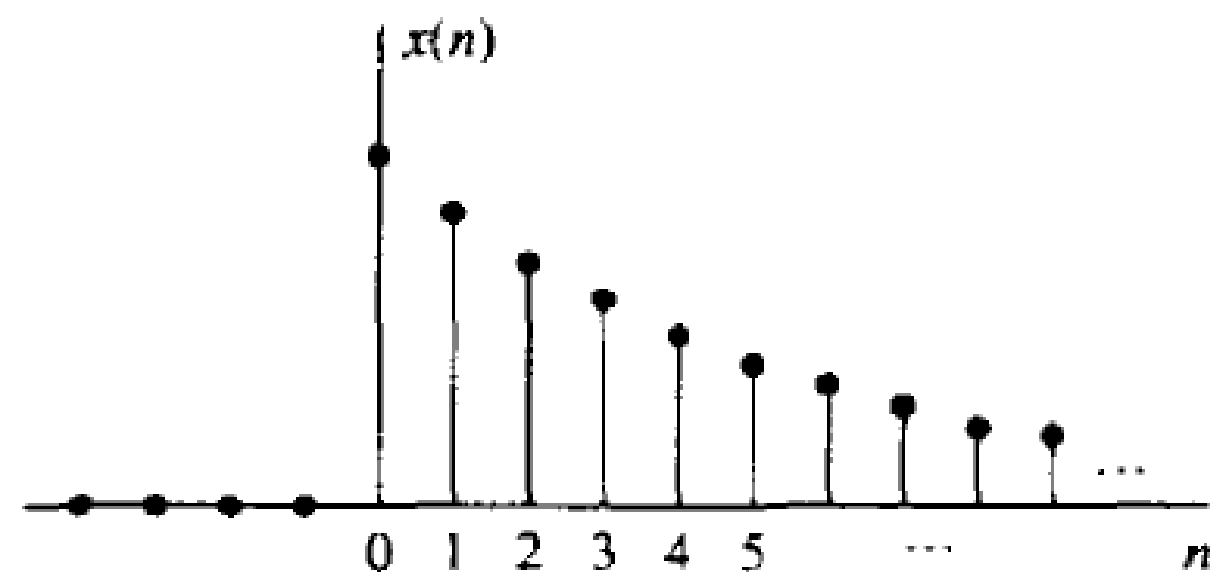


Figure 3.2 The exponential signal $x(n) = \alpha^n u(n)$ (a), and the ROC of its z -transform (b).

Example 3.1.4

Determine the z -transform of the signal

$$x(n) = -\alpha^n u(-n-1) = \begin{cases} 0, & n \geq 0 \\ -\alpha^n, & n \leq -1 \end{cases}$$

Solution From the definition (3.1.1) we have

$$X(z) = \sum_{n=-\infty}^{-1} (-\alpha^n) z^{-n} = - \sum_{l=1}^{\infty} (\alpha^{-1} z)^l$$

where $l = -n$. Using the formula

$$A + A^2 + A^3 + \dots = A(1 + A + A^2 + \dots) = \frac{A}{1 - A}$$

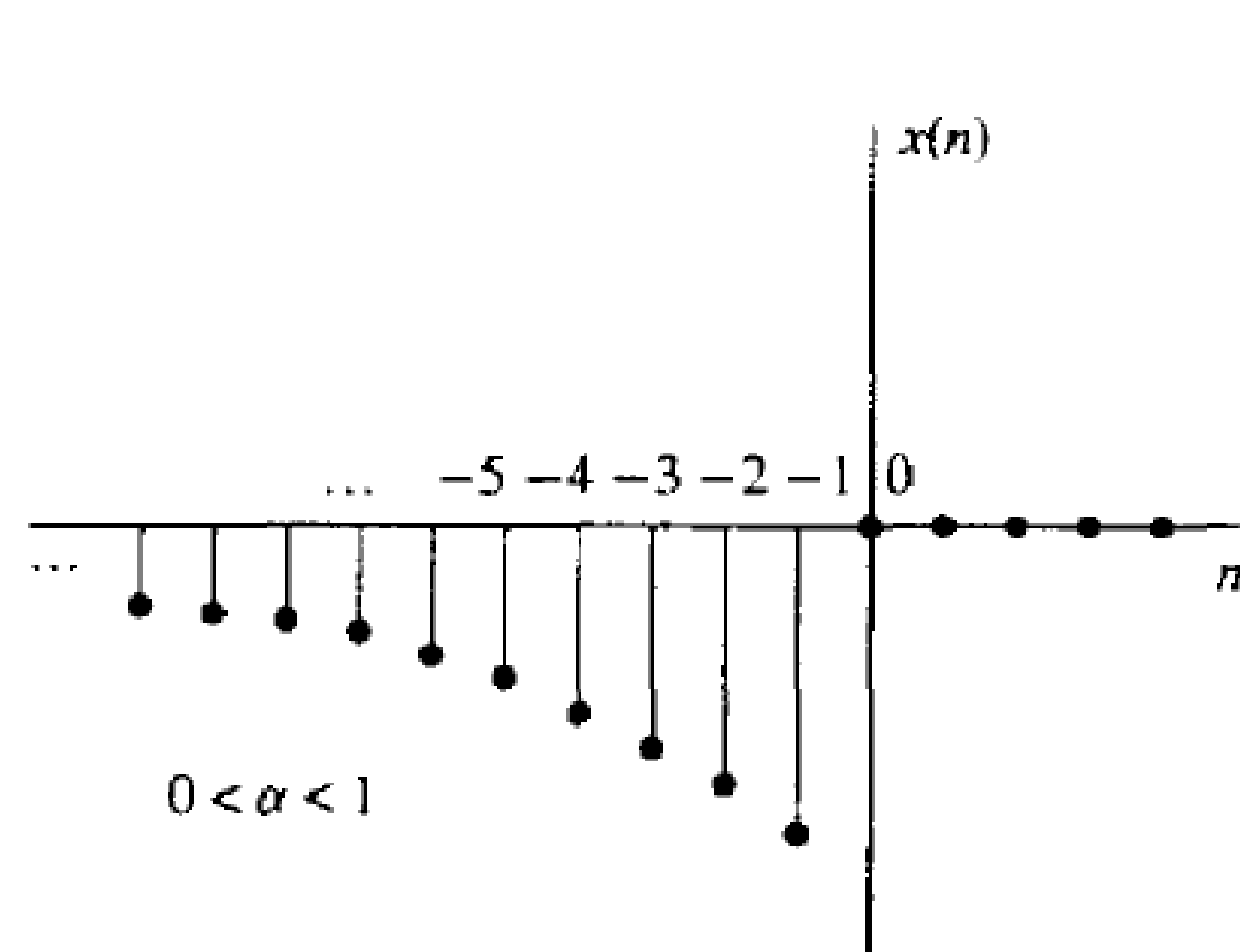
when $|A| < 1$ gives

$$X(z) = -\frac{\alpha^{-1}z}{1 - \alpha^{-1}z} = \frac{1}{1 - \alpha z^{-1}}$$

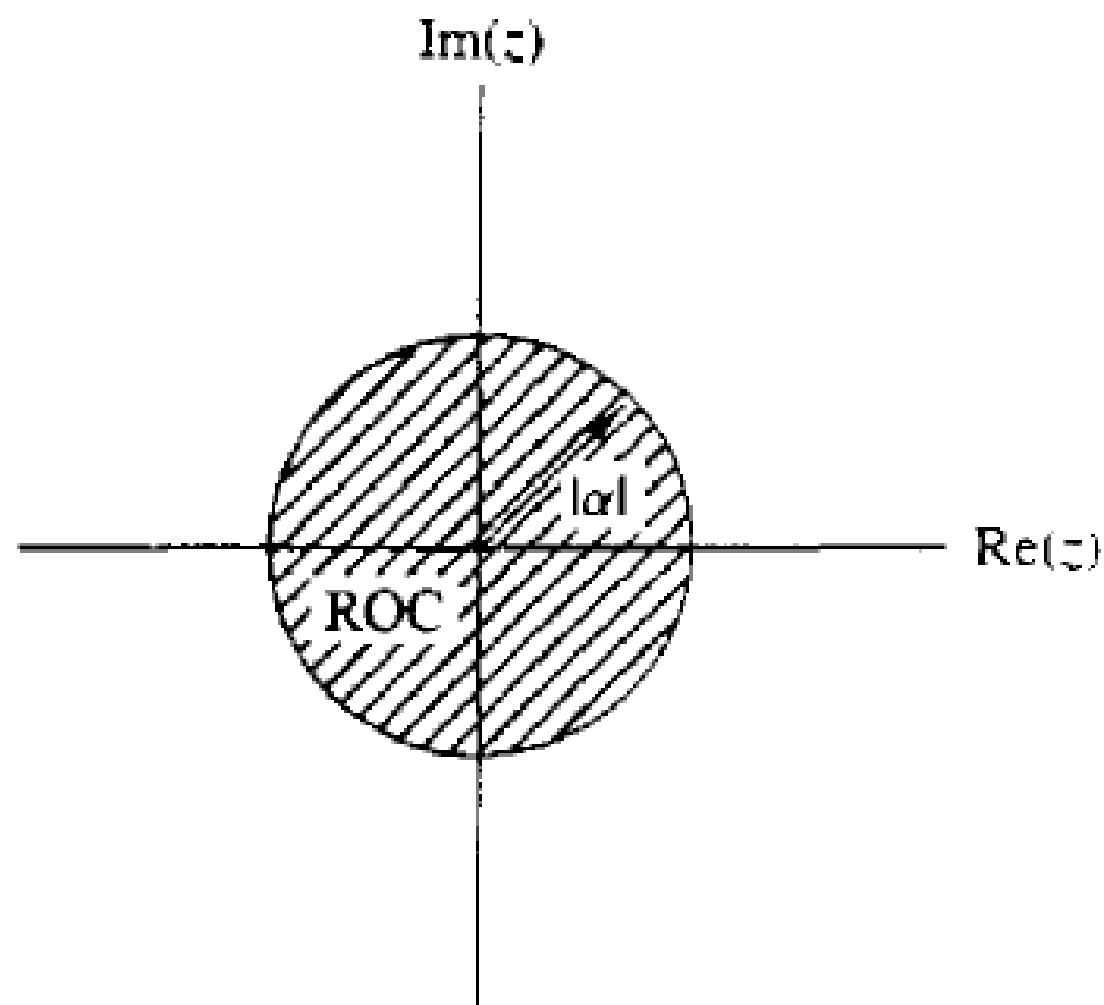
provided that $|\alpha^{-1}z| < 1$ or, equivalently, $|z| < |\alpha|$. Thus

$$x(n) = -\alpha^n u(-n-1) \xleftrightarrow{\mathcal{Z}} X(z) = -\frac{1}{1 - \alpha z^{-1}} \quad \text{ROC: } |z| < |\alpha| \quad (3.1.9)$$

The ROC is now the interior of a circle having radius $|\alpha|$. This is shown in Fig. 3.3.



(a)



(b)

Figure 3.3 Anticausal signal $x(n) = -\alpha^n u(-n-1)$ (a), and the ROC of its z -transform (b).

Examples 3.1.3 and 3.1.4 illustrate two very important issues. The first concerns the uniqueness of the z -transform. From (3.1.7) and (3.1.9) we see that the causal signal $\alpha^n u(n)$ and the anticausal signal $-\alpha^n u(-n-1)$ have identical closed-form expressions for the z -transform, that is,

$$Z\{\alpha^n u(n)\} = Z\{-\alpha^n u(-n-1)\} = \frac{1}{1 - \alpha z^{-1}}$$

This implies that a closed-form expression for the z -transform does not uniquely specify the signal in the time domain. The ambiguity can be resolved only if in addition to the closed-form expression, the ROC is specified.

In summary, a discrete-time signal $x(n)$ is uniquely determined by its z -transform $X(z)$ and the region of convergence of $X(z)$. In this text the term “ z -transform” is used to refer to both the closed-form expression and the corresponding ROC. Example 3.1.3 also illustrates the point that the ROC of a causal signal is the exterior of a circle of some radius r_2 while the ROC of an anticausal signal is the interior of a circle of some radius r_1 . The following example considers a sequence that is nonzero for $-\infty < n < \infty$.

Example 3.1.5

Determine the z -transform of the signal

$$x(n) = \alpha^n u(n) + b^n u(-n - 1)$$

Solution From definition (3.1.1) we have

$$X(z) = \sum_{n=0}^{\infty} \alpha^n z^{-n} + \sum_{n=-\infty}^{-1} b^n z^{-n} = \sum_{n=0}^{\infty} (\alpha z^{-1})^n + \sum_{l=1}^{\infty} (b^{-1} z)^l$$

The first power series converges if $|\alpha z^{-1}| < 1$ or $|z| > |\alpha|$. The second power series converges if $|b^{-1} z| < 1$ or $|z| < |b|$.

In determining the convergence of $X(z)$, we consider two different cases.

Case 1 $|b| < |\alpha|$: In this case the two ROC above do not overlap, as shown in Fig. 3.4(a). Consequently, we cannot find values of z for which both power series converge simultaneously. Clearly, in this case, $X(z)$ does not exist.

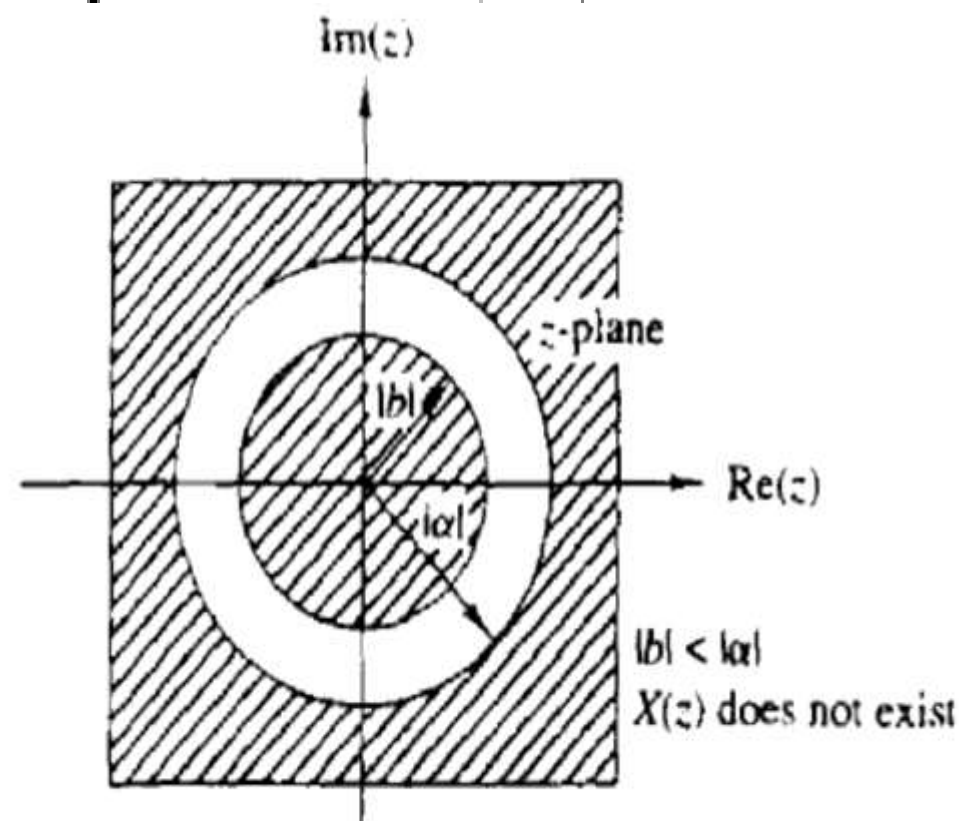


Figure 3.4 ROC for z -transform in Example 3.1.5.

Case 2 $|b| > |\alpha|$: In this case there is a ring in the z -plane where both power series converge simultaneously, as shown in Fig. 3.4(b). Then we obtain

$$\begin{aligned} X(z) &= \frac{1}{1 - \alpha z^{-1}} - \frac{1}{1 - bz^{-1}} \\ &= \frac{b - \alpha}{\alpha + b - z - \alpha b z^{-1}} \end{aligned} \quad (3.1.10)$$

The ROC of $X(z)$ is $|\alpha| < |z| < |b|$.

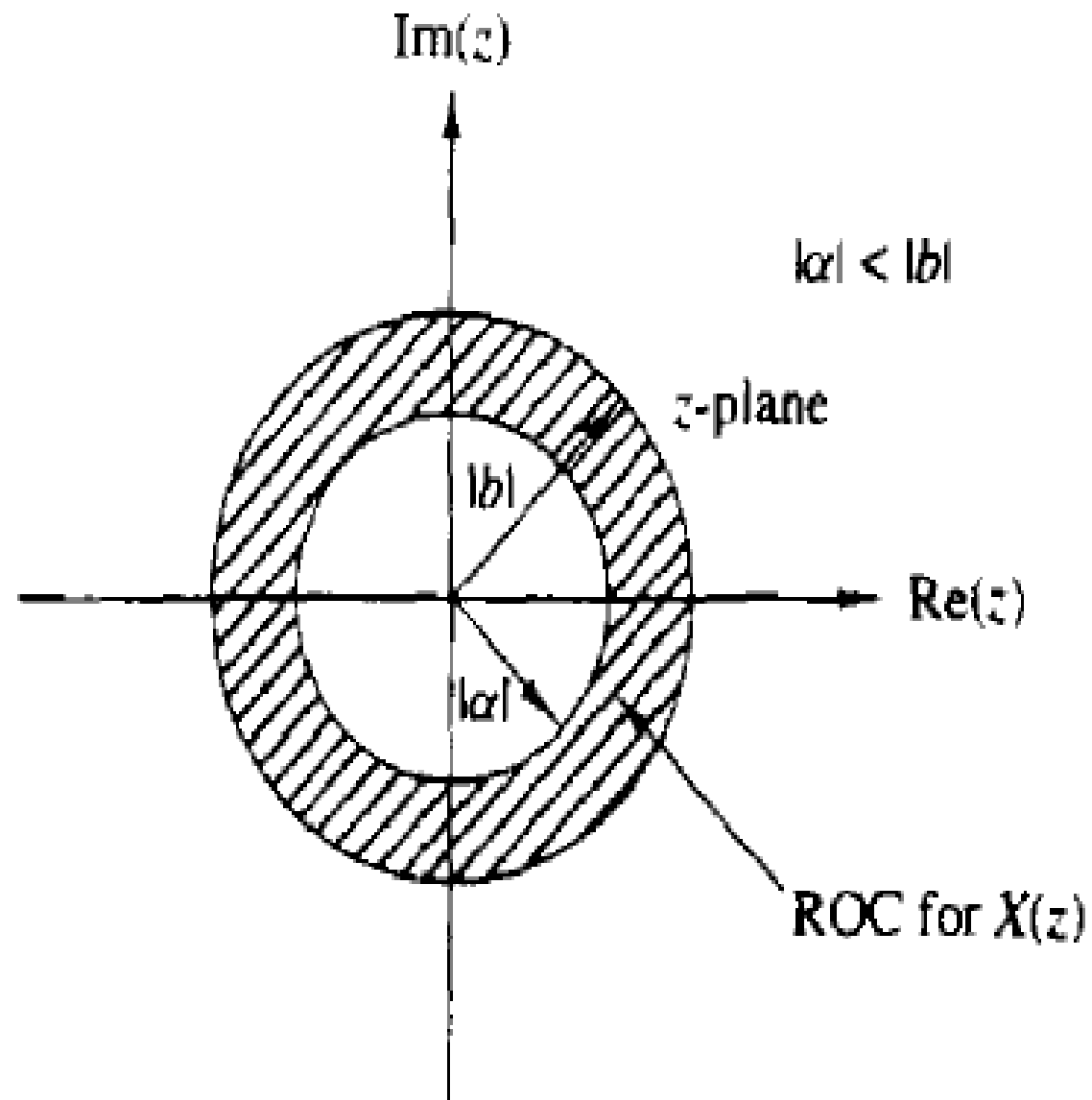


Figure 3.4 ROC for z -transform in Example 3.1.5.

Finally, we note that the z -transform defined by (3.1.1) is sometimes referred to as the *two-sided* or *bilateral* z -transform, to distinguish it from the *one-sided* or *unilateral* z -transform given by

$$X^-(z) = \sum_{n=0}^{\infty} x(n)z^{-n} \quad (3.1.11)$$

The one-sided z -transform is examined in Section 3.5. In this text we use the expression z -transform exclusively to mean the two-sided z -transform defined by (3.1.1). The term "two-sided" will be used only in cases where we want to resolve any ambiguities. Clearly, if $x(n)$ is causal [i.e., $x(n) = 0$ for $n < 0$], the one-sided and two-sided z -transforms are equivalent. In any other case, they are different.

$$X(z) \equiv \sum_{n=-\infty}^{\infty} x(n)z^{-n} \quad (3.1.1)$$

3.2 PROPERTIES OF THE Z-TRANSFORM

The z -transform is a very powerful tool for the study of discrete-time signals and systems. The power of this transform is a consequence of some very important properties that the transform possesses. In this section we examine some of these properties.

In the treatment that follows, it should be remembered that when we combine several z -transforms, the ROC of the overall transform is, at least, the intersection of the ROC of the individual transforms. This will become more apparent later, when we discuss specific examples.

Linearity. If

$$x_1(n) \xleftrightarrow{z} X_1(z)$$

and

$$x_2(n) \xleftrightarrow{z} X_2(z)$$

then

$$x(n) = a_1 x_1(n) + a_2 x_2(n) \xleftrightarrow{z} X(z) = a_1 X_1(z) + a_2 X_2(z) \quad (3.2.1)$$

for any constants a_1 and a_2 .

Example 3.2.1

Determine the z -transform and the ROC of the signal

$$x(n) = [3(2^n) - 4(3^n)]u(n)$$

Solution If we define the signals

$$x_1(n) = 2^n u(n)$$

and

$$x_2(n) = 3^n u(n)$$

then $x(n)$ can be written as

$$x(n) = 3x_1(n) - 4x_2(n)$$

According to (3.2.1), its z -transform is

$$X(z) = 3X_1(z) - 4X_2(z)$$

From (3.1.7) we recall that

$$\alpha^n u(n) \xleftrightarrow{z} \frac{1}{1 - \alpha z^{-1}} \quad \text{ROC: } |z| > |\alpha|$$

By setting $\alpha = 2$ and $\alpha = 3$ in (3.2.2), we obtain

$$x_1(n) = 2^n u(n) \xleftrightarrow{z} X_1(z) = \frac{1}{1 - 2z^{-1}} \quad \text{ROC: } |z| > 2$$

$$x_2(n) = 3^n u(n) \xleftrightarrow{z} X_2(z) = \frac{1}{1 - 3z^{-1}} \quad \text{ROC: } |z| > 3$$

The intersection of the ROC of $X_1(z)$ and $X_2(z)$ is $|z| > 3$. Thus the overall transform $X(z)$ is

$$X(z) = \frac{3}{1 - 2z^{-1}} - \frac{4}{1 - 3z^{-1}} \quad \text{ROC: } |z| > 3$$

Q. Show that $z \notin \{1\} \Rightarrow \frac{z}{z-1}, |z| > 1$

Sol:

$$X(z) = \sum_{n=0}^{\infty} x(n) z^{-n}$$

$$= \sum_{n=0}^{\infty} z^{-n}$$

$$= z^0 + z^{-1} + z^{-2} + \dots$$

$$= 1 + z^{-1} + z^{-2} + \dots = (1 - z^{-1})$$

$$= \text{scribble} = \left(1 - \frac{1}{z}\right)^{-1}$$

$$= \left(\frac{z-1}{z}\right)^{-1}$$

$$= \frac{z}{z-1}$$